Epileptic Seizure Prediction

Group 3 *- Bosio, Francesca; Devcich, Paula; Fiumara, Davide; Salgado, Valentin*

Abstract **- Epilepsy is a neurological disorder that can cause significant distress in the lives of those affected. The use of state-of-the-art technologies to prevent or anticipate seizures can substantially improve patients' safety and quality of life. In this paper, we propose a robust multimodal biocybernetic approach that leverages real-time EEG, ECG, and PPG signals to predict epileptic events in advance.**

Keywords - EEG, ECG, PPG, Epilepsy, Seizure Prediction.

I. Introduction

Epilepsy is one of the most prevalent chronic neurological disorders worldwide, affecting approximately 65 million people and representing a major health problem[1]. Its incidence is relatively uniform across high-income countries and higher in low-income regions, particularly in rural areas, and is associated with reduced life expectancy and long-term clinical and social consequences[1][2].

This disorder is characterized by a lasting predisposition of the brain to generate spontaneous epileptic seizures, accompanied by neurobiological, cognitive, psychological, and social consequences[2]. According to the International League Against Epilepsy (ILAE), epileptic seizures are transient events resulting from abnormal, excessive, and pathologically synchronous neuronal activity in the brain[1]. At the physiological level, seizures arise from pathological alterations in neuronal signaling, leading to hypersynchronous neuronal discharges that disrupt normal brain communication[3].

Clinically, epilepsy encompasses a heterogeneous spectrum of seizure types and syndromes. Seizures are primarily classified by onset as focal, generalized, or unknown, while epilepsy types are categorized as focal, generalized, combined generalized and focal, or unknown[2].

The electrophysiological abnormalities underlying epilepsy are commonly associated with an imbalance between excitatory and inhibitory activity within cortical networks, resulting in sustained hypersynchronous and oscillatory neuronal behavior[2]. These disturbances interfere with normal neuronal processing and contribute to cognitive impairment, mood disorders, and behavioral alterations, thereby affecting daily activities, autonomy, and quality of life[1]. Importantly, seizure generation is a dynamic process evolving over time through preictal (the phase preceding a seizure), ictal (the phase occurring during a seizure), postictal (the phase following a seizure), interictal (the periods between seizures), and prediction horizon (the brief interval prior to seizure onset during which alerts are required)[4].

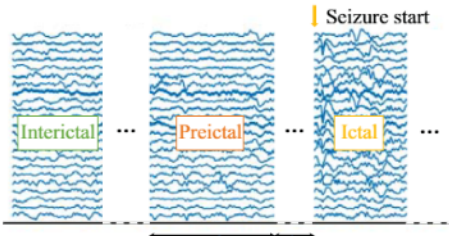

Figure 1: *Relationship of interictal, preictal, ictal and postictal*

II. Data acquisition and representation

A. Electroencephalography

Electroencephalography (EEG) records the brain's electrical activity and is widely used to study epilepsy. Scalp EEG (sEEG) is a non-invasive, cost-effective technique that offers high temporal resolution and greater safety compared to intracranial EEG. For training our model, we will use the CHB-MIT EEG dataset, which includes 844 hours of continuous sEEG recordings from 23 subjects aged from 1.5 to 19 years, collected at a sampling rate of 256 Hz[4]. Real-time data acquisition will be done with the REMI Remote EEG Monitoring System, as it is a wireless system with only 4 sensors, taking care of patients' comfortability and movement freedom[5].

B. Electrocardiography, Photoplethysmography and their junction

Electrocardiogram (ECG) and photoplethysmogram (PPG) are non-invasive physiological signals that provide complementary information on cardiovascular and autonomic function. ECG records the electrical activity of the heart, while PPG is an optical technique that detects blood volume changes in the microvascular tissue and allows the extraction of the peripheral pulse waveform[6]. Both signals are regulated by the autonomic nervous system (ANS).

Epilepsy is associated with ANS dysfunction that influences cardiac activity and peripheral hemodynamics. Cardiac and vascular dynamics can therefore be considered peripheral markers of central autonomic control, with alterations occurring not only during seizures but also during interictal and preictal phases[7]. In this context, ECG provides information on heart rate dynamics, while PPG reflects vascular responses.

Preictal increases in heart rate precede electrographic and clinical seizure onset, with higher prevalence in adults and in temporal lobe epilepsy[8]. These changes may occur seconds to minutes before seizure onset.

PPG enables the computation of pulse transit time (PTT) when combined with ECG. PTT is defined as the

time delay between the R peak of the ECG signal and the subsequent peak of the PPG waveform and is inversely related to arterial blood pressure. Since seizure activity can induce transient blood pressure variations, PTT represents an additional parameter of interest for capturing autonomic and hemodynamic changes associated with epileptic events.

In this work, ECG and PPG data were obtained in a laboratory setting, eliciting epileptic seizure under controlled conditions, using standard clinical seizure-provocation protocols (intermittent photic stimulation, hyperventilation and sleep-wake transitions)[6]. The recordings performed during controlled sessions allowed to avoid the severe class imbalance typically observed in long-term monitoring of spontaneous seizures.

For real-time acquisition we will use the Samsung Galaxy Watch4 or posterior models[9].

III. Multimodal and multiscale integration

The three signals are meant to be used in a multimodal classifier biocybernetic system that can predict in real-time the occurrence of a seizure. To do so, we use different approaches adequate for each signal.

Signal processing and classification are performed using a sliding-window approach, where EEG, ECG, and PPG signals are segmented into 20-second windows with 50% overlap, yielding a classification output every 10 seconds. This design choice balances temporal resolution with computational efficiency and ensures that critical physiological information, such as HRV and PTT intervals, is not truncated across window boundaries. A 20-second window also allows to consider information transmission delay from EEG to ECG signal[10].

For EEG signals, we will use the Dual-Octave Fusion Attention Network (DOFANet), a deep learning architecture for real-time seizure prediction because of its high performance regarding accuracy, sensitivity, specificity and f1-score. It is important to mention that the model is separated into two different blocks: one of them takes the handcrafted features, while the other the raw EEG signals.

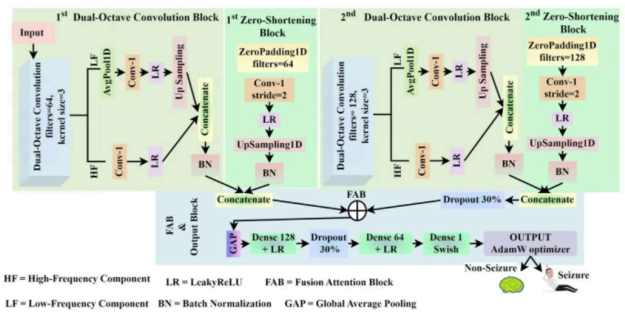

Figure 2: *DOFANet architecture*

Preictal period refers to EEG segments captured from 35 to 5 min before seizure onset. EEG signals were preprocessed to eliminate common artifacts such as those from eye blinks and muscle movement. Both in real-time application and training stages, a band-pass filter between 0.1-80 Hz and a notch filter at 50 Hz were used to clean the signals effectively. The dataset exhibits a significant class imbalance, with more non-preictal than preictal samples. To enhance representation of the preictal class, two techniques were applied: Synthetic Minority Oversampling Technique (SMOTE), to generate synthetic preictal instances by interpolating between two existing samples, and a random undersampling to non-preictal samples.

Temporal-domain features include root mean square (RMS), variance, skewness, kurtosis, zero-crossing rate (ZCR), and Teager-Kaiser energy operator (TKEO). In the spectral domain, features were derived using short-time Fourier transform (STFT). These include band power (in standard EEG frequency bands), spectral entropy, power spectral density (PSD), and mean frequency.

The Dual-Octave Convolution Block extracts multiscale temporal features by decomposing EEG signals into high- and low-frequency components, enabling independent yet complementary processing of seizure-related patterns. Additionally, the Zero-Shortening Block mitigates artifacts introduced by zero-padding in convolutional layers by discarding padding-affected boundaries and preserving only reliable signal regions. This design enhances feature integrity and improves downstream classification performance.

After the first two steps, the model employs a combination of Concatenation, Dropout, and the Fusion Attention Block (FAB) to integrate, regularize, and enhance learned features. The concatenation layer merges high- and low-frequency features along the channel axis. To reduce overfitting and enhance generalization, a Montecarlo dropout layer is applied to the concatenated features in the training phase. The Fusion Attention Block is designed to refine the extracted features by learning attention weights across temporal and channel dimensions.

Despite these advancements, seizure identification from sEEG remains challenging due to the complexity of brain signals and the subtle differences between normal and epileptic activity[4].

To complement EEG data, ECG and PPG signals are incorporated in this pipeline. To improve the quality of the signals, both for training and for the real time application, we implemented a bandpass filter from 0.4 to 20 Hz for the ECG signal and a bandpass filter from 0.4 to 4 Hz for the PPG signal[12].

A traditional machine learning pipeline is adopted, performing a comparative measurement between the most common models used in this context (e.g. Decision Tree, Support Vector Machine, Nearest Neighbor)[6]. HRV and PTT parameters are computed, including statistical descriptors (e.g., mean, variance, percentiles, skewness, kurtosis), time-domain metrics (e.g., Root Mean Square of Successive Differences), and nonlinear features such as Lorenz plot descriptors and multifractality measures[11]. These features reflect ANS activity and blood pressure variability, which are strongly associated with seizure dynamics. Feature importance is subsequently analyzed

using SHAP values, enabling the selection of the most informative features while reducing computational complexity without compromising prediction performance.

The combined use of EEG, ECG, and PPG improves robustness in real-world scenarios, particularly in outpatient settings where motion artifacts and external disturbances are common. Since bioelectrical signals (EEG and ECG) and optical signals (PPG) are affected differently by noise sources, their integration enhances signal reliability and supports more robust multimodal seizure monitoring[6].

IV. Task selection

The primary task addressed by the proposed framework is binary time-series classification, aiming to distinguish between preictal (class 1) and non-preictal (class 0) states. This is directly aligned with the objective of seizure prediction, as early identification of the preictal phase enables timely warnings and preventive interventions before seizure onset.

Decision-level fusion

The final decision is obtained through decision-level fusion, where the independent classifications derived from EEG, ECG, and PPG signals are combined. This strategy is particularly suitable for wearable systems, as it allows the framework to remain operational even if one modality becomes temporarily unavailable due to sensor detachment, noise, or signal loss. It therefore enhances both reliability and practicality in real-world use.

Alarm generation follows a hierarchical scheme based on the combined classification outputs. When both classifiers predict class 0, no alarm state is issued. If the class predicted by both classifiers is 1, a decision-time period is set up: for the alarm to be triggered, class 1 must be obtained in all windows inside the time period. If a preictal state is detected by only one classificator, uncertainty quantification is computed using Monte Carlo dropout layers on the deep learning EEG model. The dropout layer is applied only in disagreement cases to reduce the computational cost. The classification output is given by a weighted prediction (wp):

$$wp = W_{EEG} * score_{EEG} + W_{ECG,PPG} * score_{ECG,PPG}$$

Where the W_{EEG} is given by 1 - variance of the Monte Carlo dropout output, and the $W_{ECG,PPG}$ by 1 - W_{EEG}.

Stronger alerts, including audio alarms and safety messages, are generated when both classificators indicate preictal activity for more than a threshold number of consequent windows in preictal state. This graduated response reduces false alarms while ensuring timely warnings for high-risk situations.

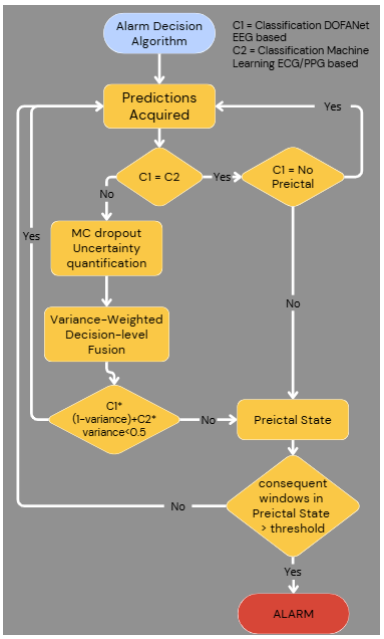

***Figure 3**: Alarm decision algorithm flow chart*

Previous studies demonstrate that each classifier is independently capable of predicting epileptic seizures. EEG-based models have shown that preictal states can be reliably identified using EEG signals alone, achieving prediction accuracies above 85%[6]. Similarly, ECG-based seizure prediction relying on cardiac and autonomic features has achieved comparable performance, with accuracies around 85-86% using ECG signals alone[11]. These results indicate that both modalities are individually predictive, providing a rationale for their combination through decision-level fusion.

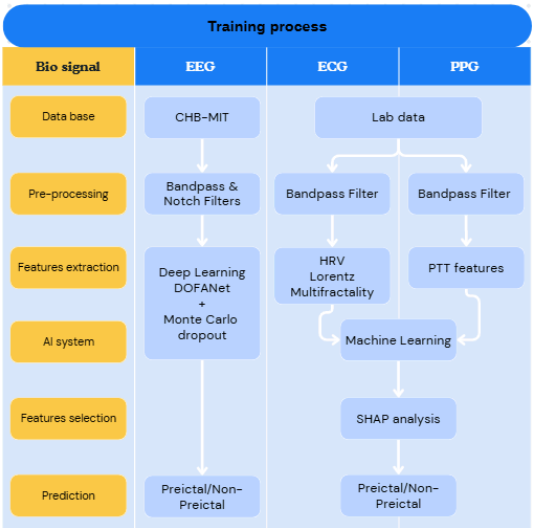

***Figure 4:** Training process diagram*

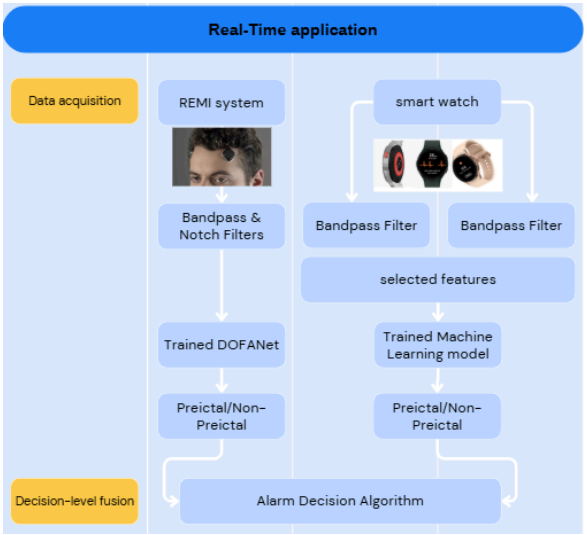

Figure 5: *Real-Time application diagram*

V. Biocybernetic loop

Once trained, the proposed system operates within a broader biocybernetic loop that actively involves the patient, clinician, and bioengineer, enabling continuous personalization and long-term optimization of seizure management. This closed-loop structure ensures that model predictions are not static but evolve in response to real-world performance and individual patient needs.

To support clinical decision-making, the system periodically generates summary reports for the clinician, including the number of alarms issued, and detected preictal events. The reporting interval, which may range from a few days to several weeks, is tailored by the clinician based on the patient's condition and monitoring requirements. These reports facilitate therapy adjustments, clinical interpretation of seizure patterns, and evaluation of the system's effectiveness over time.

In parallel, technical performance reports are delivered to the bioengineer, focusing on classification accuracy, false positive and false negative reported by the patient. This feedback enables systematic tuning of decision thresholds and fusion rules, supporting patient-specific calibration and improving the trade-off between sensitivity and specificity. Such adaptive optimization is essential for deploying a personalized wearable device in real-world settings.

At the patient level, the biocybernetic loop provides immediate and actionable feedback through visual and audio alarms delivered via the smartwatch interface. These alerts allow the patient to take preventive actions, such as moving to a safe environment or adopting a protective posture before seizure onset, which represents the primary functional goal of the device. To further strengthen patient engagement and transparency, the framework includes a regular self-reporting mechanism, such as a weekly questionnaire, through which patients can provide feedback on perceived accuracy, usability, satisfaction, and impact on quality of life. This subjective feedback is shared with both clinicians and bioengineers, closing the loop between algorithmic performance, clinical oversight, and patient experience.

By integrating explainable AI methods, multimodal decision fusion, and structured human-in-the-loop feedback, the proposed biocybernetic framework supports adaptive, transparent, and patient-centered epilepsy management. This continuous interaction between biological signals, intelligent algorithms, and human stakeholders enhances trust, personalization, and clinical relevance in seizure prediction systems.

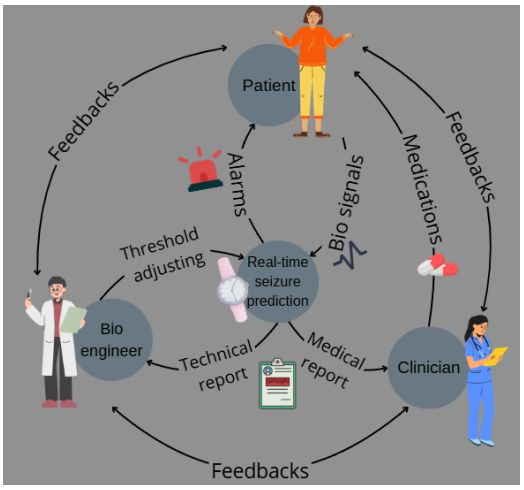

Figure 6: *Biocybernetic loop*

REFERENCES

[1] S. L. Moshé, E. Perucca, P. Ryvlin, and T. Tomson, "Epilepsy: New advances," *The Lancet*, vol. 385, no. 9971, pp. 884–898, Feb. 28, 2015. [Online]. Available: https://doi.org/10.1016/S0140-6736(14)60456-6

[2] R. D. Thijs, R. Surges, T. J. O'Brien, and J. W. Sander, "Epilepsy in adults," *The Lancet*, vol. 393, no. 10172, pp. 689–701, Feb. 16, 2019. [Online]. Available: https://doi.org/10.1016/S0140-6736(18)32596-0

[3] National Institute of Neurological Disorders and Stroke, "Epilepsy and seizures," Apr. 7, 2025. [Online]. Available: https://www.ninds.nih.gov/health-information/disorders/epilepsy-and-seizures

[4] P. K. Upadhyay, P. Kumar, and S. K. Mishra, "Efficient EEG-based seizure prediction using DOFANet: A deep learning approach," *International Journal of Information Technology*, vol. 17, pp. 5121–5134, 2025. [Online]. Available: https://doi.org/10.1007/s41870-025-02646-2

[5] Epitel Inc., "Epitel," [Online]. Available: https://www.epitel.com/

[6] M. Author *et al.*, "EEG-based seizure detection and prediction," *Sensors*, vol. 22, no. 23, p. 9372, Nov. 2022. [Online]. Available: https://www.mdpi.com/1424-8220/22/23/9372

[7] A. B. Author *et al.*, "Title of the article," *Epilepsia*, vol. 50, no. 9, pp. 2135–2143, Sep. 2009. [Online]. Available: https://doi.org/10.1111/j.1528-1167.2009.02479.x

[8] A. B. Author *et al.*, "Title of the article," *Seizure*, vol. 53, pp. 10–17, 2017. [Online]. Available: https://www.sciencedirect.com/science/article/pii/S1059131117307343

[9] Samsung Electronics Co., Ltd., "Samsung Health Sensor Overview," *Samsung Developer*, 2025. [Online]. Available: https://developer.samsung.com/health/sensor/overview.html

[10] Devinsky O. "Effects of Seizures on Autonomic and Cardiovascular Function". Epilepsy Curr. 2004 Mar;4(2):43-46. [online]. Available: https://pmc.ncbi.nlm.nih.gov/articles/PMC531654/

[11] A. Abtahi *et al.*, "Identification of relevant ECG features for epileptic seizure prediction," in *Proceedings of IEEE*, 2025. [Online]. Available: https://ieeexplore.ieee.org/document/11052221

[12] D. Zambrana-Vinaroz, J. M. Vicente-Samper, and J. M. Sabater-Navarro, "Validation of continuous monitoring system for epileptic users in outpatient settings," *Sensors*, vol. 22, no. 8, p. 2900, Apr. 9, 2022. [Online]. Available: https://www.mdpi.com/1424-8220/22/8/2900